Bina.az Housing Price Prediction

From-scratch decision trees and Pegasos SVM

PegaSold

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MLE-AI-201 Final Project V3 · National AI Center

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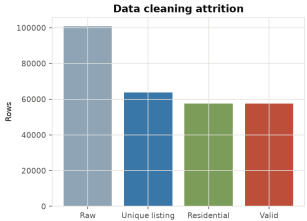
Project question — Speaker 1: data and task

- Real Bina.az asking-price scrape: 100,775 raw rows from October–November 2024.
- Task A predicts $\ln(\mathrm{price})$; we report log-space and AZN-space errors.
- Task B classifies standard vs premium; the training median cutoff is 208,000 AZN.
- The decision tree and Pegasos are NumPy implementations; sklearn is used as a comparison reference.

Cleaning and leakage — Speaker 1

- Keep the latest scrape per normalized advert ID; retain Baku residential listings.
- Apply declared price, area, AZN/m^2, room and floor checks.
- Train-only log-price MAD fence: 26,130–1,671,651 AZN; removed 246 train rows only.
- Split: 40,033/8,632/8,632; validation/test rows remain intact.
- Exclude target-derived prices, free text, identifiers, seller fields and raw address.
- Fit imputation, scaling and category vocabulary from training rows.





Exploration and feature design — Speaker 1

- Parse Azerbaijani measurements; encode flags as 0/1.
- Use area, rooms, floor, land, coordinates, category and frequent locations.
- Engineer distance to centre, floor ratio, area per room and land ratio.
- Spatial and property-type patterns provide signal, not causal proof.
- This is a historical asking-price scrape, not transaction data.

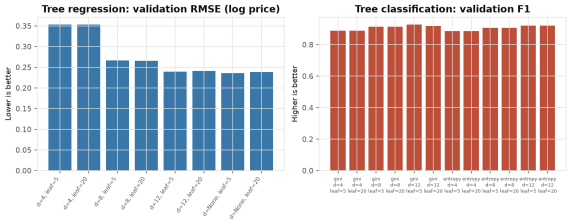


Split protocol and cross-validation — Speaker 1

- Fixed seed; split allocation balances price bins and keeps repeated advert IDs together.
- Premium threshold is computed from training prices only: 208,000 AZN.
- Validation chooses hyperparameters; the held-out test is used after selection.
- Five-fold CV uses all 40,033 training rows (about 32k fit / 8k validation per fold).
- Preprocessing statistics were fitted before folds; CV is indicative, not fully nested.

Decision tree from scratch — Speaker 2

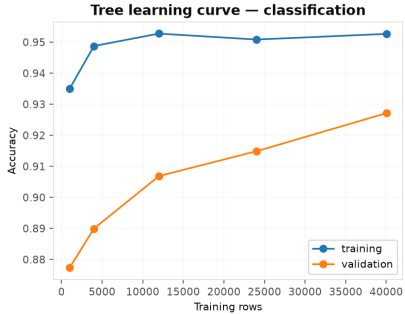
- Sort numeric values and test boundaries between distinct adjacent values.
- Classification: Gini/entropy information gain; regression: MSE reduction.
- Cumulative counts and sums make split scoring efficient.
- Leaves predict majority class or mean target.
- Depth and minimum-leaf limits control model variance.
- Validation selects criterion, depth and leaf size; test remains untouched.



Tree results and generalization — Speaker 2

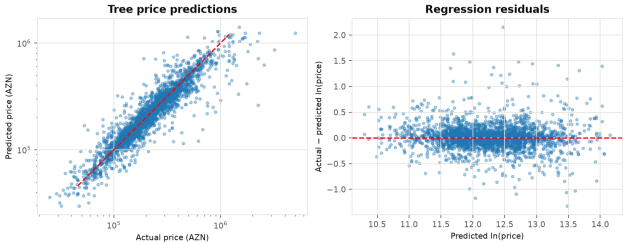
Metric	Score	Metric	Score
Task A log RMSE	0.239	Task B accuracy	0.922
Task A AZN RMSE	147,356	Task B F1 / AUC	0.923 / 0.958
Reference log RMSE	0.239	Reference accuracy	0.922





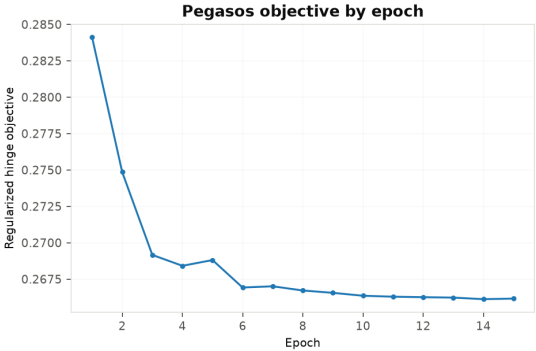
Regression error in AZN — Speaker 2

- Held-out test MAE: 47,613 AZN; median absolute error: 16,431 AZN.
- Test RMSE: 147,356 AZN.
- Largest absolute error: 3,760,714 AZN.
- Actual 5,000,000 AZN; predicted 1,239,286 AZN.
- Test retains extreme prices outside the training MAD fence.



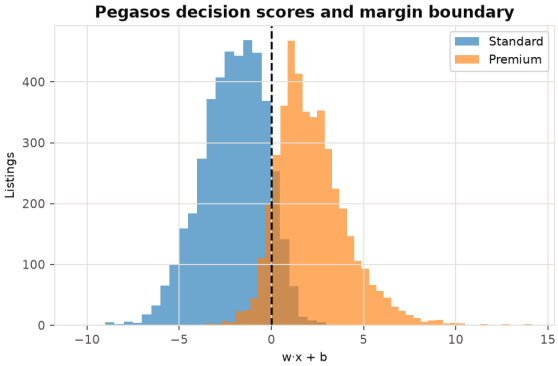
Linear Pegasos SVM — Speaker 3

- Minimize regularized hinge loss with stochastic sub-gradient updates.
- Standardize using training data; labels map to $-1, +1$.
- Selected $\lambda = 0.01$; margin width $2/\|w\| = 0.900$.
- Support vectors (signed margin ≤ 1): 11,587 training rows.
- Test accuracy 0.902; F1 0.903; ROC-AUC 0.968.



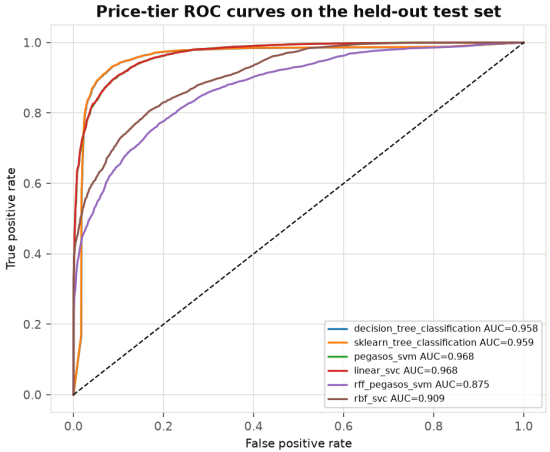
Nonlinear SVM and decision margin — Speaker 3

- Seeded Random Fourier Features approximate an RBF kernel; the NumPy Pegasos optimizer trains the mapped features.
- 128 components; selected $\gamma = 0.01$; training subset 8,000.
- Matched RBF SVC uses the same subset; results depend on random features and seed.
- Nonlinear mapping did not outperform the linear model in this run.



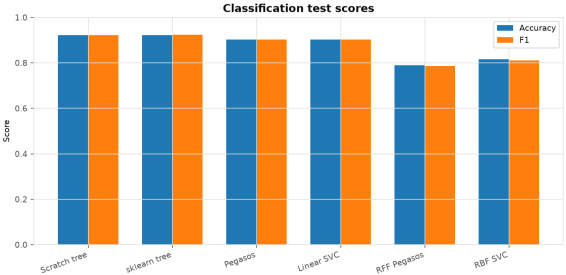
Fair comparison and uncertainty — Speaker 4

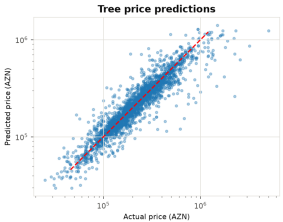
- Tree references use matched criteria, depth and leaf settings.
- LinearSVC uses hinge loss, fold-local scaling and objective-matched $C = 1/(\lambda n)$.
- Five-fold CV covers all 40,033 training rows.
- Regression tree CV MAE: 41,505 $\pm$ 675 AZN.
- sklearn tree CV MAE: 41,516 $\pm$ 696 AZN.
- CV MAE is lower partly because filtered folds do not contain the test's full extreme-price tail.

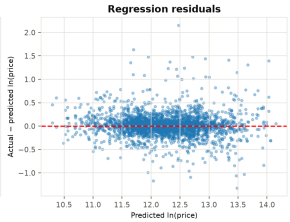


Held-out test comparison — Speaker 4

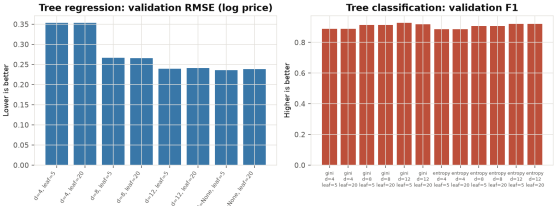
Model	Accuracy	F1	AUC	Fit sec
Scratch tree	0.922	0.923	0.958	1.76
sklearn tree	0.922	0.923	0.959	0.24
Pegasos	0.902	0.903	0.968	2.81
LinearSVC	0.902	0.903	0.968	0.48
RFF Pegasos	0.790	0.786	0.875	0.82



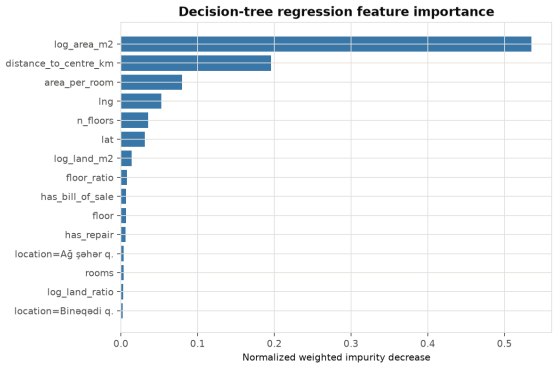




Tuning and feature importance — Speaker 4



Validation sweeps.



Impurity importance is descriptive, not causal.

Conclusions and limitations — All

- Price-tier classification is easier than exact advertised-price prediction; luxury-tail errors remain large.
- Scratch and sklearn trees are close under matched settings.
- The held-out test MAE is 47,613 AZN; CV mean does not replace this result.
- Asking prices are not verified transactions; one 2024 scrape is not today's market.
- CV preprocessing precedes folds; ambiguous cross-ID duplicates may remain.

Reproducibility and handoff — All

1. Speaker 1: data preparation, EDA, feature design and split protocol.
2. Speaker 2: NumPy tree, validation tuning and regression error analysis.
3. Speaker 3: Pegasos optimization, margin and nonlinear RFF model.
4. Speaker 4: sklearn benchmarks, CV, metrics and reproducibility.

Reproduce

`python -m src.run_all` `pytest -q`
Results, figures and caveats: `reports/final_report.md`.

AI assistance was used for coding suggestions, debugging, corrections and review; the team checked the final code and results.